

FR-01 Tiered Tariff Calculation Engine

FR-01	Tiered Tariff Calculation Engine
Primary Actor(s)	Homeowner, Tenant, Guest
Stakeholders and Interest	Interested in understanding billing based on consumption tiers
Trigger	User requests a cost breakdown or usage report.
Pre-conditions	The system has recorded consumption data in Resource Usage.
Post-conditions	Total cost is calculated using peak Rate and operate
Main Success Scenario	1. User selects "View Cost Prediction". 2. Analytics Engine calls Tariff Calculator. 3. System calculates cost based on tiered rates. 4. Result is displayed to the user
Extensions	-----
Priority	High
Special Requirements	Calculations must sync with real-time utility price updates.

FR-02 View Carbon Footprint

FR-02	View Carbon Footprint
Primary Actor(s)	Homeowner
Stakeholders and Interest	Interested in monitoring environmental impact.
Trigger	User selects the carbon footprint option on the dashboard.
Pre-conditions	Resource Usage data for electricity, water, and gas is available.
Post-conditions	An estimated CO2 value is generated.
Main Success Scenario	1. User triggers "View Carbon Footprint". 2. Analytics Engine retrieves data from Resource Usage. 3. System converts resource units to carbon metrics. 4. Report is displayed.
Extensions	-----
Priority	Medium
Special Requirements	----

FR-03 View Cost Prediction

FR-03	View Cost Prediction
Primary Actor(s)	Homeowner
Stakeholders and Interest	Needs to forecast monthly expenses to manage budgets.
Trigger	Periodic system update or user manual refresh
Pre-conditions	Sufficient data exists in usage History.
Post-conditions	A predicted bill amount is stored and displayed.
Main Success Scenario	1. Analytics Engine processes usage History. 2. Algorithm predicts future usage based on trends. 3. System displays the "View Cost Prediction" output.
Extensions	Budget Overrun Alert: Triggered if prediction exceeds the set budget.
Priority	High
Special Requirements	Must account for seasonal consumption variations.

FR-04 Compare Usage / Historical Fata Analysis

FR-04	Compare Usage / Historical Data Analysis
Primary Actor(s)	Homeowner
Stakeholders and Interest	System: Needs baselines to identify "Vampire Power" or anomalies.
Trigger	Completion of a data collection cycle (e.g., 30 days).
Pre-conditions	Device usage history is populated in Appliance Signature.
Post-conditions	Baseline consumption patterns are established.
Main Success Scenario	1. System analyzes Appliance Signature and usage History. 2. Analytics Engine generates a standard usage profile. 3. User can now "Compare Usage" against this baseline.
Extensions	-----
Priority	Medium
Special Requirements	-----

FR-05 Manage Appliances / View Resource Usage

FR-05	Manage Appliances / View Resource Usage
Primary Actor(s)	System (Internal Logic)
Stakeholders and Interest	User: Needs data in readable units (e.g., kWh to Cost).
Trigger	Data display request involving different measurement units.
Pre-conditions	Unit Converter class is initialized.
Post-conditions	Values are accurately converted and displayed.
Main Success Scenario	1. System fetches raw data from Sensor Simulator. 2. Unit Converter applies convert Energy or convert Currency. 3. Converted data is passed to Report Service.
Extensions	-----
Priority	Medium
Special Requirements	Must support various international unit standards.

FR-06 Detect Anomalies / Emergency Shutdown

FR-06	Detect Anomalies / Emergency Shutdown
Primary Actor(s)	System
Stakeholders and Interest	Homeowner: Interested in safety and preventing appliance damage.
Trigger	Sensor Simulator detects data outside normal thresholds.
Pre-conditions	AlertRule and Alert Engine are active.
Post-conditions	Notification sent; system may initiate "Emergency Shutdown".
Main Success Scenario	1. Sensor Simulator sends a simulated Anomaly signal. 2. Alert Engine identifies the breach of an AlertRule. 3. System triggers the "Detect Anomalies" use case. 4. "Maintenance Alert" is generated for the user.
Extensions	Receive Notifications: Alert is pushed to the user's device.
Priority	Critical
Special Requirements	Anomaly detection must function with minimal latency.

FR-07 Solar / Renewable Offset Tracker

FR-07	Solar / Renewable Offset Tracker
Primary Actor(s)	<i>Homeowner</i>
Stakeholders and Interest	<i>User, System Admin</i>
Trigger	<i>User opens renewable energy dashboard</i>
Pre-conditions	• <i>User has an account</i> • <i>Energy consumption data available</i>
Post-conditions	<i>Renewable offset is calculated and displayed</i>
Main Success Scenario	1. User opens dashboard 2. System retrieves energy consumption data 3. System retrieves renewable production data 4. System calculates offset percentage 5. Results displayed in charts
Extensions	No renewable data → show “No Data Available”
Priority	<i>High</i>
Special Requirements	• <i>Real-time update</i> • <i>Graph visualization</i>

FR-08 Comparative Neighborhood Benchmarking

FR-08	Comparative Neighborhood Benchmarking
Primary Actor(s)	<i>Homeowner</i>
Stakeholders and Interest	<i>Homeowner, System</i>
Trigger	<i>User selects comparison feature</i>
Pre-conditions	• <i>User location defined</i> • <i>Neighborhood data exists</i>
Post-conditions	<i>Comparison report displayed</i>
Main Success Scenario	1. User selects benchmarking option 2. System retrieves neighborhood data 3. System compares usage 4. Displays ranking and insights
Extensions	2a: No neighborhood data → show fallback message
Priority	<i>Medium</i>
Special Requirements	• <i>Data anonymization</i> • <i>Fast response time</i>

FR-09 Historical Data Aggregator

FR-09	Historical Data Aggregator
Primary Actor(s)	<i>Homeowner</i>
Stakeholders and Interest	<i>Homeowner</i>
Trigger	<i>User requests historical report</i>
Pre-conditions	<i>Stored historical data available</i>
Post-conditions	<i>Aggregated data displayed</i>
Main Success Scenario	1. User selects date range 2. System fetches historical data 3. System aggregates data 4. Displays charts
Extensions	2a: Missing data → partial display
Priority	<i>High</i>
Special Requirements	• <i>Support large datasets</i> • <i>Smooth graph rendering</i>

FR-10 Resource-to-Task Translator

FR-10	Resource-to-Task Translator
Primary Actor(s)	<i>Homeowner</i>
Stakeholders and Interest	<i>Homeowner</i>
Trigger	<i>User views consumption details</i>
Pre-conditions	<i>Resource usage calculated</i>
Post-conditions	<i>Usage converted into real-life tasks</i>
Main Success Scenario	1. System receives usage data 2. Converts it into equivalent tasks (e.g., hours of AC usage) 3. Displays results
Extensions	2a: Unknown mapping → show generic message
Priority	<i>Medium</i>
Special Requirements	• <i>Clear, simple language</i> • <i>Accurate mapping</i>

FR-11 Weather-Correlation Engine

FR-11	Weather-Correlation Engine
Primary Actor(s)	<i>System</i>
Stakeholders and Interest	<i>Homeowner, Tenant</i>
Trigger	<i>System detects new weather data</i>
Pre-conditions	• <i>Weather API connected</i> • <i>Consumption data exists</i>
Post-conditions	<i>Correlation insights generated</i>
Main Success Scenario	1. System retrieves weather data 2. Matches with consumption data 3. Calculations correlation 4. Displays insights
Extensions	1a: API failure → use cached data
Priority	<i>High</i>
Special Requirements	• <i>External API integration</i> • <i>Data accuracy</i>

FR-12 Appliance Signature Library

FR-12	Appliance Signature Library
Primary Actor(s)	<i>System, Homeowner</i>
Stakeholders and Interest	<i>Homeowner, Tenant</i>
Trigger	<i>New appliance data added or analyzed</i>
Pre-conditions	<i>Appliance database exists</i>
Post-conditions	<i>Appliance patterns stored/updated</i>
Main Success Scenario	1. System collects appliance usage patterns 2. Identifies unique signatures 3. Stores in library 4. Uses data for future predictions
Extensions	2a: Unknown appliance → mark as unidentified
Priority	<i>Medium</i>
Special Requirements	• <i>Machine Learning Support</i> • <i>Scalable storage</i>

FR-13 Health Degradation Simulator

FR-13	Health Degradation Simulator
Primary Actor(s)	<i>System</i>
Stakeholders and Interest	<i>Stakeholders: Homeowner, Maintenance Technicians, Device Manufactures</i> <i>Interest: Monitor appliance health over time, predict potential failures before they occur, Improve appliance lifespan and efficiency</i>
Trigger	<i>Periodic system update</i> <i>New appliance usage data is recorded</i> <i>Detection of abnormal operating conditions</i>
Pre-conditions	<i>Appliance is registered in the system.</i> <i>Usage data is available.</i>
Post-conditions	<i>Appliance health scores are updated.</i>
Main Success Scenario	<i>1. The system collects appliance usage data.</i> <i>2. The system analyzes operating patterns.</i> <i>3. The system calculates the current health score.</i> <i>4. The system updates the appliance health status.</i> <i>5. The updated health information is stored and displayed.</i>
Extensions	<i>1. Insufficient usage data: The system uses default degradation models.</i> <i>2. Abnormal conditions detected: The system accelerates health degradation calculation.</i>
Priority	<i>High</i>
Special Requirements	<i>1.The system shall support configurable degradation models.</i> <i>2.The system shall handle large volumes of historical usage data efficiently.</i> <i>3.The system shall ensure high accuracy in health score calculations.</i>

FR-14 Maintenance Alert Trigger

FR-14	Maintenance Alert Trigger
Primary Actor(s)	<i>System/User</i>
Stakeholders and Interest	<i>Homeowners → avoid breakdowns</i> <i>Maintenance Technicians → faster response</i>
Trigger	<i>Health drops below threshold</i>
Pre-conditions	<i>1.Appliance health monitoring is active.</i> <i>2.Threshold values are defined.</i>
Post-conditions	<i>1.Maintenance alert is generated.</i> <i>2.Notification is sent to the user.</i> <i>3.Alert is logged in the system.</i>
Main Success Scenario	<i>1. The system monitors appliance health.</i> <i>2. The system detects low health.</i> <i>3. The system generates an alert.</i> <i>4. The system notifies the user.</i> <i>5. The alert is logged.</i>
Extensions	<i>1. Slight drop → warning instead of alert</i> <i>2.Notification failure → retry</i>
Priority	<i>High</i>
Special Requirements	<i>1. Real-time alerts</i> <i>2.Multiple notification channels</i> <i>3.Custom thresholds.</i>

FR-15 Smart-Plug Association Logic

FR-15	Smart-Plug Association Logic
Primary Actor(s)	<i>User</i>
Stakeholders and Interest	<i>Homeowner → control & monitoring</i> <i>Household Members → ease of use</i> <i>Manufacturers → integration quality</i>
Trigger	<i>User initiates pairing</i>
Pre-conditions	<i>1. Appliance is registered.</i> <i>2. Smart plug device is available and powered on.</i>
Post-conditions	<i>1. Smart plug is successfully linked to the appliance.</i> <i>2. Energy data collection begins.</i> <i>3. User can control the appliance remotely.</i>
Main Success Scenario	<i>1. User selects appliance.</i> <i>2. User starts pairing.</i> <i>3. System detects smart plugs.</i> <i>4. System links plugs to appliances.</i> <i>5. System starts collecting data.</i> <i>6. User controls devices remotely.</i>
Extensions	<i>1. Plug not detected → retry</i> <i>2. Association fails → error message</i>
Priority	<i>High</i>
Special Requirements	<i>1. Support multiple plug types</i> <i>2. Secure pairing</i> <i>3. Reliable communication.</i>

FR-16 Appliance Lifecycle Tracker

FR-16	Appliance Lifecycle Tracker
Primary Actor(s)	<i>System/User</i>
Stakeholders and Interest	<i>Homeowner → replacement planning</i> <i>Technicians → maintenance planning</i> <i>Manufacturers → lifecycle insights</i>
Trigger	<i>Appliance added</i> <i>Usage updated</i>
Pre-conditions	<i>1. Appliance is added to the system.</i> <i>2. Initial data is available.</i>
Post-conditions	<i>1. Lifecycle data is updated.</i> <i>2. Remaining lifespan is estimated.</i> <i>3. Notifications are triggered if needed.</i>
Main Success Scenario	<i>1. System records installation data.</i> <i>2. System tracks usage.</i> <i>3. System calculates lifecycle.</i> <i>4. System estimates remaining life.</i> <i>5. System updates status.</i> <i>6. System notifies user if needed.</i>
Extensions	<i>1. Missing data → use averages</i> <i>2. Uncertain estimate → show range</i>
Priority	<i>Medium</i>
Special Requirements	<i>1. Long-term data storage</i> <i>2. Configurable lifecycle models</i> <i>3. Clear visualization</i>

FR-17 Duty Cycle Monitor

FR-17	Duty Cycle Monitor
Primary Actor(s)	<i>System</i>
Stakeholders and Interest	<i>Homeowner → usage awareness</i> <i>Manufacturers → efficiency improvement</i>
Trigger	<i>Device ON/OFF change</i>
Pre-conditions	<i>1. Appliance is connected and monitored.</i> <i>2. ON/OFF state data is available.</i>
Post-conditions	<i>1. Duty cycle is calculated.</i> <i>2. Usage patterns are analyzed and stored.</i>
Main Success Scenario	<i>1. System monitors ON/OFF states.</i> <i>2. System records cycles.</i> <i>3. System calculates duty cycle.</i> <i>4. System analyzes patterns.</i> <i>5. System stores results.</i>
Extensions	<i>1. Incomplete data → estimation</i> <i>2. Abnormal cycles → flag issue</i>
Priority	<i>Medium</i>
Special Requirements	<i>1. High-frequency sampling</i> <i>2. Accurate state detection</i> <i>3. Robust data handling</i>

FR-18 Standby Power Management

FR-18	Standby Power Management
Primary Actor(s)	<i>System/User</i>
Stakeholders and Interest	<i>Homeowners → reduce bills</i> <i>Environmental Organizations → sustainability</i> <i>Utility Providers → efficient energy use</i>
Trigger	<i>Device enters standby</i> <i>Low power detected</i>
Pre-conditions	<i>1. Device is connected to the system.</i> <i>2. Power consumption data is available.</i>
Post-conditions	<i>1. Standby devices are identified.</i> <i>2. Recommendations or actions are applied.</i> <i>3. Energy usage records are updated.</i>
Main Success Scenario	<i>1. System detects standby devices.</i> <i>2. System measures consumption.</i> <i>3. System analyzes waste.</i> <i>4. System provides recommendations.</i> <i>5. System may turn off devices.</i> <i>6. System updates records.</i>
Extensions	<i>1. Not sure standby → continue monitoring</i> <i>2. Auto-off disabled → recommend only</i>
Priority	<i>High</i>
Special Requirements	<i>1. High detection accuracy</i> <i>2. User control over automation</i> <i>3. Minimize false positives</i>

19 Manage Appliance Permissions

FR-19	Appliance Permission Levels
Primary Actor(s)	<i>Homeowner</i>
Stakeholders and Interest	• <i>Homeowner: Full control over appliance access.</i> • <i>Tenant/Guest: Limited and controlled access.</i>
Trigger	<i>The user chooses to modify appliance permissions</i>
Pre-conditions	• <i>User is logged in</i> • <i>Appliance is registered in the system</i>
Post-conditions	• <i>Permissions are successfully updated</i>
Main Success Scenario	1. <i>User navigates to Manage Users & Roles.</i> 2. <i>Select an appliance.</i> 3. <i>Assigns permissions (On/Off / View Only / Full Control).</i> 4. <i>Saves change.</i>
Extensions	• <i>3a: Invalid permission selection → Error message displayed.</i>
Priority	<i>High</i>
Special Requirements	• <i>Support multiple user roles</i> • <i>Simple and intuitive UI</i>

20 Simulate Firmware Update

FR-20	Device Firmware Update Simulator
Primary Actor(s)	<i>System / Homeowner</i>
Stakeholders and Interest	• <i>User: Ensure update safety before execution</i>
Trigger	<i>User requests a firmware update</i>
Pre-conditions	• <i>Device is connected</i> • <i>Firmware update is available</i>
Post-conditions	• <i>Permissions are successfully updated</i>
Main Success Scenario	1. <i>User initiates firmware update</i> 2. <i>System runs simulation</i> 3. <i>Displays results (Success / Risk warnings)</i> 4. <i>User decides whether to proceed</i>
Extensions	• <i>2a: Simulation fails → Warning displayed</i>
Priority	<i>Medium</i>

Special Requirements	Simulation must not affect the real device
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21 View Interaction Logs

FR-21	Appliance Interaction Log
Primary Actor(s)	<i>Homeowner</i>
Stakeholders and Interest	<i>User: Monitor and track appliance usage</i>
Trigger	<i>User requests to view logs</i>
Pre-conditions	Log data exists
Post-conditions	<i>Logs are displayed</i>
Main Success Scenario	- User opens logs section - System displays activities (On/Off/Changes) - User filters log by date or user
Extensions	No logs available → Informational message
Priority	<i>Medium</i>
Special Requirements	Long-term data storage

22 Emergency Shutdown>

FR-22	Emergency Shutdown Protocol
Primary Actor(s)	<i>System / Homeowner</i>
Stakeholders and Interest	- User: Safety and protection - System: Prevent damage
Trigger	<i>An anomaly is detected or manual activation</i>
Pre-conditions	An anomaly is detected or manual activation
Post-conditions	All appliances are shut down
Main Success Scenario	- System detects anomaly - Sends alert notification - Executes automatic or manual shutdown - Logs the event
Extensions	3a: Shutdown fails → Critical alert sent
Priority	<i>Critical</i>
Special Requirements	Real-time response

23 Automatic Budget Adjustment>

FR-19	Dynamic Budget Adjuster
Primary Actor(s)	<i>System</i>
Stakeholders and Interest	<i>User: Reduce energy costs</i>
Trigger	<i>Budget threshold is exceeded</i>
Pre-conditions	<i>Budget is set</i>
Post-conditions	<i>Energy usage is adjusted</i>
Main Success Scenario	<i>System monitors energy usage</i> <i>Detects budget overrun</i> <i>Adjusts appliance usage or suggests actions</i> <i>Updates budget status</i>
Extensions	<i>User rejects adjustment → No changes applied</i>
Priority	<i>High</i>
Special Requirements	<i>Integration with cost prediction</i>

24 Participate in Eco Challenge

FR-24	Eco-Challenge Engine
Primary Actor(s)	<i>Homeowner / Tenant</i>
Stakeholders and Interest	<i>User: Save energy and earn rewards</i>
Trigger	<i>User joins a challenge</i>
Pre-conditions	<i>User is registered</i> <i>Challenge is available</i>
Post-conditions	<i>Progress and rewards are updated</i>
Main Success Scenario	<i>User joins a challenge</i> <i>System tracks energy usage</i> <i>Compare usage against goals</i> <i>Rewards points are granted</i>
Extensions	<i>Goal not achieved → No reward</i>

Priority	<i>Medium</i>
Special Requirements	• <i>Gamification (points, rewards system)</i>

FR-31: Budget Overrun Escalation

FR-31	Budget Overrun Escalation
Primary Actor(s)	Homeowner
Stakeholders and Interest	Homeowner: Wants to stay within budget limits and receive timely notifications. System: Needs to monitor usage and trigger alerts at defined thresholds.
Trigger	Resource usage reaches 80%, 90%, or 100% of the set budget.
Pre-conditions	The Homeowner has set a budget using the 'Set Budget' use case. The system is actively monitoring resource usage.
Post-conditions	The Homeowner receives a staged notification corresponding to the threshold reached.
Main Success Scenario	1. The system continuously monitors the current resource usage against the set budget. 2. When usage reaches 80%, the system sends a first-level warning notification to the Homeowner. 3. When usage reaches 90%, the system sends a second-level alert with a stronger warning. 4. When usage reaches 100%, the system sends a critical alert indicating the budget has been fully consumed. 5. Each notification includes current usage, remaining budget, and a recommendation.
Extensions	3a. If the Homeowner has not set a budget, the system prompts them to set one. 4a. If notification delivery fails (e.g., no internet), the system queues the alert and retries. 5a. The Homeowner may choose to update the budget from within the notification.
Priority	High
Special Requirements	Notifications must be sent in real time. All three threshold levels must be tracked

	independently.
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FR-32: Vacation Mode Logic

FR-32	Vacation Mode Logic
Primary Actor(s)	Homeowner
Stakeholders and Interest	Homeowner: Wants to minimize energy waste and receive alerts for any activity while away. System: Must enforce ultra-low thresholds during the vacation period.
Trigger	The Homeowner activates 'Vacation Mode' via the settings panel.
Pre-conditions	The Homeowner is authenticated. At least one device is registered in the system.
Post-conditions	The system switches to ultra-low thresholds and flags any usage activity as an anomaly.
Main Success Scenario	1. The Homeowner navigates to the settings and selects 'Vacation Mode'. 2. The system sets the house status to 'Empty' and applies ultra-low usage thresholds for all resources. 3. Any detected usage above the near-zero threshold triggers an immediate alert. 4. The system logs all activity that occurs during Vacation Mode. 5. When the Homeowner deactivates Vacation Mode, the system restores the previous thresholds.
Extensions	3a. If an anomaly is detected, the system sends a real-time push notification and email alert. 5a. If the Homeowner forgets to deactivate, they can set an auto-expiry date for Vacation Mode.
Priority	Medium
Special Requirements	The Homeowner must be able to activate/deactivate Vacation Mode with a single action.

FR-33: Rule-Based Logic Engine

FR-33	Rule-Based Logic Engine
Primary Actor(s)	Homeowner
Stakeholders and Interest	Homeowner: Wants to automate responses to resource conditions. System: Must evaluate and execute user-defined 'If-This-Then-That' rules.
Trigger	The Homeowner creates or edits an automation rule via 'Create Automation Rule'.
Pre-conditions	The Homeowner is authenticated. The sensor data stream is active.
Post-conditions	The rule is saved and begins executing whenever its conditions are met.
Main Success Scenario	1. The Homeowner opens the Rule Builder interface. 2. The Homeowner defines an 'If' condition (e.g., If Gas > X and Temp > Y). 3. The Homeowner defines a 'Then' action (e.g., Send Alert, Trigger Shutdown). 4. The Homeowner saves the rule. The system validates the rule syntax. 5. The system continuously evaluates all active rules against incoming sensor data. 6. When conditions are met, the system executes the defined action and logs the event.
Extensions	4a. If the rule has invalid syntax, the system displays an error and requests correction. 6a. If the action fails (e.g., device offline), the system logs the failure and notifies the Homeowner.
Priority	High
Special Requirements	The engine must support multiple conditions (AND/OR logic). Rules must be persisted across sessions.

FR-34: Sensor Data Mock-Generator

FR-34	Sensor Data Mock-Generator
Primary Actor(s)	System
Stakeholders and Interest	Homeowner / Tenant: Benefit from realistic data for testing and demonstration. System: Must produce realistic data without relying on physical hardware.
Trigger	The system starts up or a user requests a data simulation session.
Pre-conditions	The system is running. No physical sensors are required.
Post-conditions	Randomized but realistic resource data points are available in the system every few seconds.
Main Success Scenario	1. The system initializes the Sensor Data Mock-Generator on startup. 2. The generator produces random data points for electricity, water, and gas usage. 3. Data values follow realistic patterns (e.g., higher usage in morning/evening). 4. The data generated is fed into the system's Resource Usage records every few seconds. 5. All dashboards, charts, and alerts respond to the mock data as if it were real.
Extensions	3a. The generator can simulate anomalies (e.g., sudden spikes) if the 'simulate Anomaly' flag is set. 5a. The administrator can pause or reset the generator from the system settings.
Priority	High
Special Requirements	Data must be statistically realistic. The generator must not block the main application thread.

FR-35: Multi-User Role-Based Access Control (RBAC)

FR-35	Multi-User Role-Based Access Control (RBAC)
Primary Actor(s)	Homeowner
Stakeholders and Interest	Homeowner: Needs full administrative control over the system. Tenant: Needs limited access to their own usage data. Guest: Needs read-only access to basic information. System: Must enforce permissions consistently across all features.
Trigger	A user logs in or the Homeowner assigns/modifies a role via 'Manage Users & Roles'.
Pre-conditions	The Homeowner is authenticated. The user account exists in the system.
Post-conditions	The user's session is governed by the permissions defined for their role.
Main Success Scenario	1. The Homeowner navigates to 'Manage Users & Roles'. 2. The Homeowner assigns a role (Homeowner, Tenant) to a registered user. 3. The system updates the user's role and associated permissions. 4. When the user logs in, the system checks their role and restricts UI and API access accordingly. 5. The Homeowner can modify or revoke roles at any time.
Extensions	4a. A Tenant attempting to access a restricted feature receives an 'Access Denied' message. 5a. If a user's role is changed while they are active, the system applies the new permissions on their next action.
Priority	High
Special Requirements	Permissions must be enforced at both UI and backend levels. Role changes must be logged in the System Audit Trail.

FR-36: Notification Channel Manager

FR-36	Notification Channel Manager
Primary Actor(s)	System, Homeowner
Stakeholders and Interest	Homeowner: Wants to receive alerts through their preferred channels. System: Must route alerts to the correct channel based on priority.
Trigger	An alert is triggered by any monitoring function (budget overrun, anomaly detection, maintenance, etc.).
Pre-conditions	The user's notification preferences are configured. At least one alert has been triggered.
Post-conditions	The alert is delivered through the highest-priority available channel.
Main Success Scenario	1. An alert event is generated by the system (e.g., Budget Overrun, Anomaly Detected). 2. The system reads the user's notification channel preferences and the alert's priority level. 3. Based on priority, the system selects the appropriate channel: Dashboard (low), Email (medium), or SMS (high). 4. The system delivers the notification through the selected channel. 5. The system logs the notification delivery status (sent, failed, queued).
Extensions	4a. If the primary channel fails, the system falls back to the next available channel. 5a. If all channels fail, the system queues for the notification and retries when connectivity is restored.
Priority	High
Special Requirements	Channel selection logic must be configurable per user. SMS is simulated in the current version.

FR-37: System Self-Diagnostic

FR-37	System Self-Diagnostic
Primary Actor(s)	System
Stakeholders and Interest	Homeowner: Interested in system stability and receiving alerts if any hardware or software component fails.
Trigger	Scheduled interval (e.g., every 60 minutes) or system boot-up.
Pre-conditions	The system is powered on and operational.
Post-conditions	System status log is updated; maintenance alerts are triggered if issues are found.
Main Success Scenario	1. System initiates a check of internal communication modules. 2. System verifies connectivity with sensors and appliances. 3. System checks database integrity and available storage 4. System records a "Healthy" status in the diagnostic log.
Extensions	Component Failure: System triggers a Maintenance Alert and sends a notification to the user.
Priority	High
Special Requirements	Diagnostics must run in the background without interrupting real-time monitoring.

FR-38: Data Export / Report Builder

FR-38	Data Export / Report Builder
Primary Actor(s)	Homeowner
Stakeholders and Interest	Homeowner: Needs to extract energy data for external analysis or financial documentation.
Trigger	User selects the "Generate Reports" option.
Pre-conditions	User is authenticated (Login) and has data available in "Historical Data Analysis".
Post-conditions	A downloadable file (PDF/CSV) is generated and provided to the user.
Main Success Scenario	1. User selects "Generate Reports". 2. User filters data by date range or resource type (e.g., Energy/Water). 3. System executes "Export Reports" logic. 4. System compiles the data into the requested format 5. User downloads the file.
Extensions	No Data Found: System notifies the user that no records exist for the selected period.
Priority	Medium
Special Requirements	Exported files must be compatible with standard spreadsheet software (e.g., Excel).

FR-39: Time-Sync Synchronization

FR-39	Time-Sync Synchronization
Primary Actor(s)	System
Stakeholders and Interest	System: Requires precise timing for "Create Automation Rule" and accurate timestamping of "Historical Data".
Trigger	Periodic (every 24 hours) or upon reconnection to the internet.
Pre-conditions	Internet connectivity is available.
Post-conditions	System clock is synchronized with a global NTP (Network Time Protocol) server.
Main Success Scenario	1. System requests at current time from a reliable NTP server. 2. System receives time packet and calculates network latency. 3. System updates internal clock to match global time.
Extensions	Connection Lost: System relies on internal RTC (Real-Time Clock) until the next successful sync.
Priority	High
Special Requirements	Clock drift must not exceed 1 second between synchronization intervals.

FR-40: Encryption Wrapper

FR-40	Encryption Wrapper
Primary Actor(s)	System
Stakeholders and Interest	All Users: Ensure privacy of resource usage data and personal account information.
Trigger	Data is sent to the database or transmitted over the network.
Pre-conditions	Security keys are properly initialized in the system.
Post-conditions	Data is stored or transmitted in an encrypted, non-readable format.
Main Success Scenario	1. System captures raw data (e.g., Login credentials or Resource Usage). 2. System applies encryption algorithm (e.g., AES-256) to the data. 3. Encrypted data is passed to the storage or transmission module.
Extensions	Key Failure: If encryption keys are missing, the system halts the transaction and logs a security error.
Priority	High
Special Requirements	Encryption must comply with industry standards for IoT data security.

FR-41: System Audit Trail

FR-41	System Audit Trail
Primary Actor(s)	System / Homeowner
Stakeholders and Interest	Homeowner: Needs to track who changed settings like "Set Budget" or "Manage Users & Roles".
Trigger	Any administrative action or configuration change.
Pre-conditions	System is active and logging is enabled.
Post-conditions	A permanent, read-only record of the action is saved.
Main Success Scenario	1. User performs an action (e.g., "Manage Appliances"). 2. System captures User ID, Timestamp, IP Address, and the specific change made. 3. System writes this data to the Audit Log.
Extensions	Storage Full: System archives oldest logs to make room for new audit entries.
Priority	Medium
Special Requirements	Audit logs must be tamper-proof and accessible only by the primary Homeowner.

FR-42: Database Purge & Archive Logic

FR-42	Database Purge & Archive Logic
Primary Actor(s)	System
Stakeholders and Interest	System: To maintain high performance for "View Resource Usage" and "Historical Data Analysis".
Trigger	Database size reaches a predefined threshold or a specific age (e.g., data older than 2 years).
Pre-conditions	Backup/Archive storage is available.
Post-conditions	Old data is moved to archive; active database size is reduced.
Main Success Scenario	1. System identifies records exceeding the retention period. 2. System copies these records to long-term archive storage. 3. System verifies archive integrity. 4. System deletes the original records from the active database.
Extensions	Archive Failure: System cancels the deletion process and triggers a "Maintenance Alert".
Priority	Low
Special Requirements	Purge operations must run during "off-peak" hours to avoid slowing down the user interface.